FINAL CAPSTONE PROJECT

Guidelines & Evaluation Rubric

Machine Learning & Data analysis Track

To ensure that our training programs provide practical, industry-oriented experience, every course will conclude with a Capstone Project that reflects the skills and knowledge acquired throughout the program.

1 MANDATORY REQUIREMENTS

- The project must solve a real-world problem or address a meaningful use case whenever possible.
- Projects may be completed in groups of up to five members.
- Students should apply the concepts learned during the course, including Data Analysis, Feature Engineering, and Classical and/or Advanced Machine Learning techniques (e.g., regression, classification, clustering, ensemble methods, or deep learning).
- The final solution must include a local deployment using either Streamlit or Flask.
- The complete project must be uploaded to GitHub with clean code organization and comprehensive documentation.
- Each student must publish their project on LinkedIn to showcase their work professionally and support future career opportunities.

CERTIFICATE REQUIREMENT

Completing, documenting, and publishing the project is a mandatory requirement for receiving the course completion certificate.

2 REQUIRED DELIVERABLES

- Public GitHub repository containing the complete source code.
- README file with project description, setup instructions, dependencies, and usage guidelines.
- Presentation slides for the final discussion session.
- A working Streamlit or Flask application.
- LinkedIn post

3 PROJECT EXPECTATIONS

- Clearly define the problem statement and its importance.
- Perform data collection, cleaning, preprocessing, and exploratory data analysis when applicable.

- Develop, train, and evaluate suitable Machine Learning models.
- Justify model choices and evaluation metrics.
- Build a simple, user-friendly local deployment interface.
- Write clean, modular, and well-documented code.
- Discuss limitations and potential future improvements.

4 FINAL PRESENTATION & DISCUSSION DAY

The final discussion session will begin at **9:00am on Thursday [30-07-2026]**. Each team must deliver a presentation and perform a live demonstration of their project.

- Problem statement and motivation.
- Dataset description and preprocessing pipeline.
- Methodology and technical approach.
- Machine Learning techniques used (e.g., feature engineering, model selection, hyperparameter tuning, ensembling).
- Model evaluation metrics and results.
- System architecture and deployment process.
- Challenges encountered and solutions implemented.
- Future work and possible improvements.
- Live demonstration of the final application.

Group of 4	Group of 5
35 min	**45 min**

5 EVALUATION RUBRIC · 100 POINTS

Criterion	Description	Points
Problem Definition	Clarity, relevance, and real-world value of the problem.	10
Data Analysis & Preprocessing	Quality of EDA, preprocessing, and data understanding.	15
Model Development	Appropriate use and evaluation of Machine Learning methods.	20
Deployment	Functionality and usability of the Streamlit or Flask application.	15
Code Quality & Documentation	Organization, readability, modularity, and GitHub documentation.	15
Presentation & Communication	Quality of slides, explanations, and responses to questions.	15
Live Demo	Successful demonstration of the final product.	10

6 FINAL NOTE

Students are encouraged to be creative and integrate multiple concepts covered during the course. The Capstone Project should serve as a portfolio-quality project that demonstrates both technical and communication skills.